Batch first, verify natively: GPU tools for orbital compute

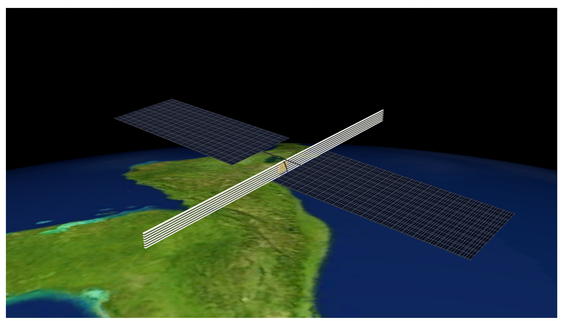
Neil Dave | Reproducible engineering study | Draft for review

8.49 million designs/s; 33.1× warm sweep speedup vs one GB10 Cortex-X925 CPU core.

Problem / contribution: GPU tools close the 30 × 250 kW / 100-t mass-and-thermal design point, then verify it natively. Batch physics, held-out checks and mechanical screens deliver a measured, reproducible engineering result—not flight qualification.

Application | requirements

30 × 250 kW = 7.5 MW installed peak
175 kW nominal (sunlit); annual 164.4 kW/sc
92.674 t incl. 2-t dispenser; 7.326 t reserve
140 panels; 861.75 m² gross array
GPU/HBM 366.16 / 361.84 K nominal
Peak inlet 391.91 K [BOL; ε = 0.92]



Earth: NASA Blue Marble; rendered in Isaac Sim (RTX path tracing); illustrative

What we designed

861.75 m² solar array.
Al 6061-T6 radiator:
7 round 8/7-mm tubes.
MPG40; safe-mode
pack: 31.25 kg.
Eclipse compute pause.
Trunk Whipple shields.
30-envelope stow.
Rigid Newton; local FEA.

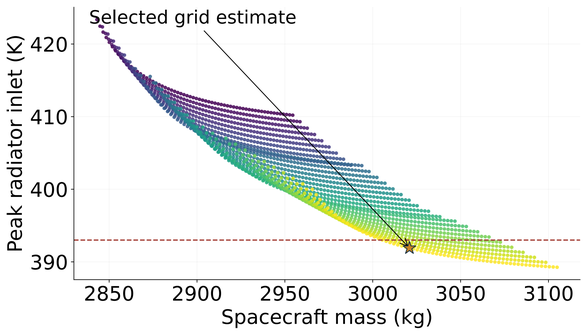
Not designed / OPEN

Header MMOD:
694 expected / 5 yr.
EOL: −16.53 kW/sc.
Full-peak GPU: 404.0 K.
Flight stow / hinges /
hoses; launcher loads.
Flexible dynamics.
TID/SEE; ADCS; links.
Spares; braze; disposal.

23-mm seams: 1.495% gross area. 319.7 W/m² gross implies 324.55 W/m² active (cell-fill OPEN). Freeze reference FAIL waived; full mass charged. Annual 175-kW interpretation OPEN.

1 Warp | discrete sizing

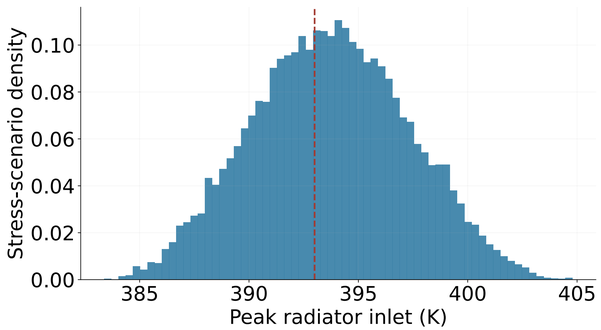
1,806,336 designs



Minimum full mass in verified stow subset. Star: surrogate estimate; native check: 3022.48 kg/sc, 140 panels.

2 Warp | stress uncertainty

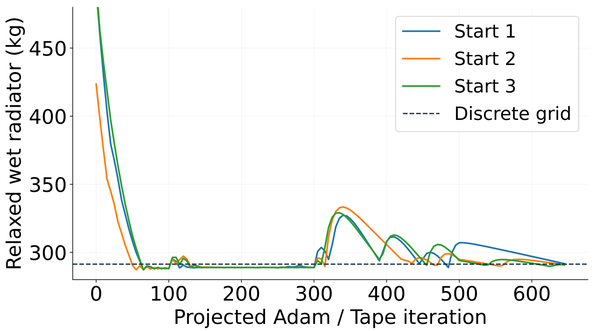
42.08% meet inlet cap



1,048,576 bounded engineering scenarios + rank sensitivity, not measured reliability. Coating spec: ε = 0.92 (BOL).

3 warp.Tape | gradients

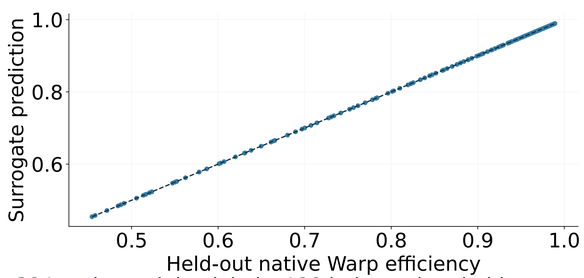
0.82% relaxed mass gap



Finite-difference gradient check PASS. Continuous relaxation cross-checks the grid; not adopted hardware or a global optimum.

4 PhysicsNeMo | held out

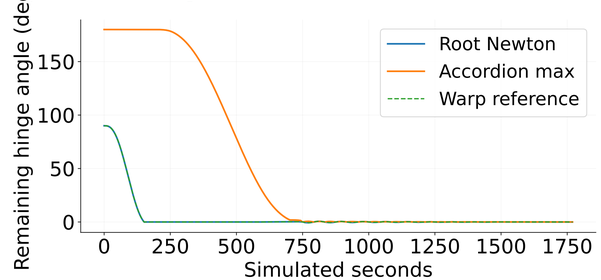
Max |Δη| = 9.3e-05



384 native training labels; 128 independent held-out labels. Selected fins re-solved natively.

5 Newton | rigid deployment

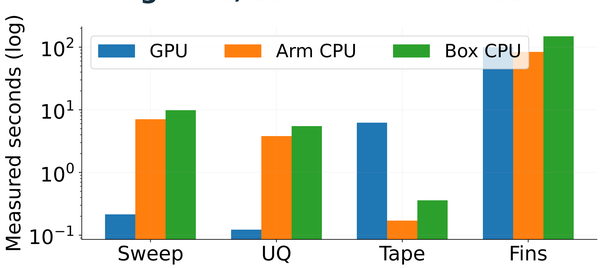
31 / 31 representative locks



Sampled clearance PASS; angle parity 0.0061 deg. Force-step convergence and flexible hardware OPEN. Both solvers on CUDA.

6 Honest speed | same fp64 work

Batching wins; serial work can lose



Same fp64 kernel; CPU: 1 core / 1 thread, not all 20 Arm cores. Warm runs exclude setup/JIT/training; serial GPU work can lose.

Hardware: GB10 GPU; CPU: 1 Cortex-X925 core / 1 thread; box: x86 VM. Warp 1.18.0 / 1.19.0; Newton 1.7.0; PhysicsNeMo 2.2.0; CalculiX 2.21.
References: SpaceX Starmind; Warp; Newton; PhysicsNeMo; CoolProp; Dow DOWFROST; Christiansen (NASA NTRS 20030068423). Links and evidence: bundle.
AI assistance: Grok Bot; the author is responsible for assumptions and claims. Engineering screens, not flight qualification.
Canonical SHA256: f02f82201c45c9b1c98b1052f8adc795f47c441e642059bd450b288b5de88846